
CIS 678 - Machine Learning

– Naive Bayes –

From Probability Estimation to Classification

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Last Class

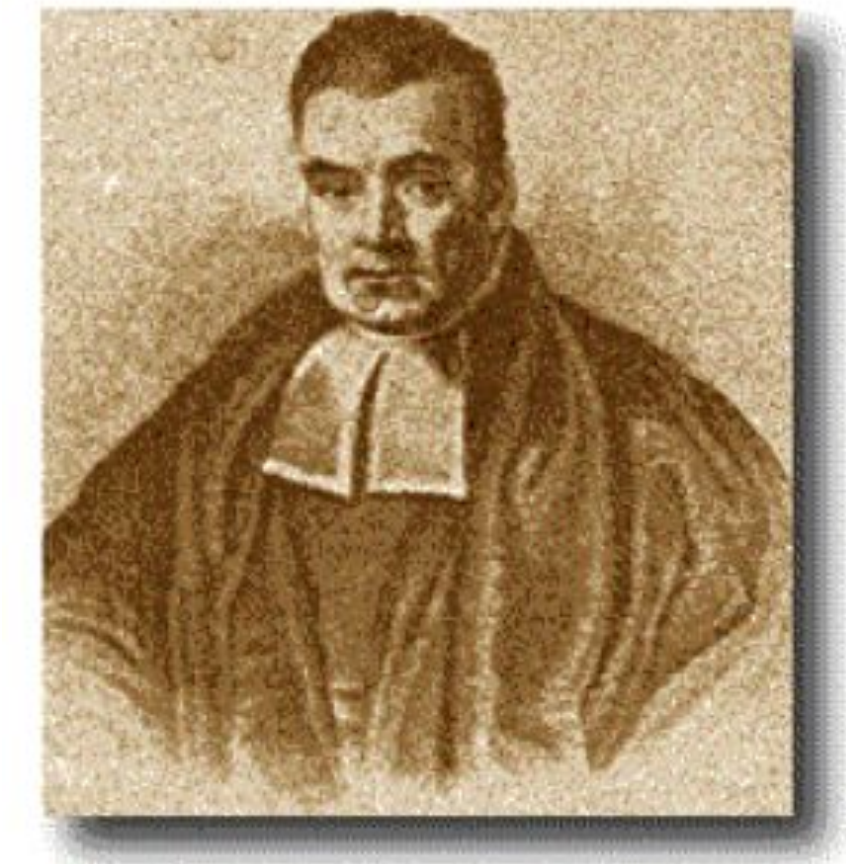
- Joint and conditional probability
- Probability versus likelihood
- MLE: estimate parameters from observed data
- MAP: combine the likelihood with prior information

Agenda

- Probabilistic classification using Bayes' rule
- Why full joint distributions do not scale
- The conditional-independence assumption
- Learning parameters with MLE and MAP
- Smoothing

Bayes' Rule

$$\underbrace{P(Y | X)}_{\text{posterior}} = \frac{\overbrace{P(X | Y)}^{\text{likelihood}} \overbrace{P(Y)}^{\text{prior}}}{\underbrace{P(X)}_{\text{evidence}}}$$



Bayes, Thomas (1763) An essay towards solving a problem in the doctrine of chances. *Philosophical Transactions of the Royal Society of London*, 53:370-418

Applying Bayes Rule

Bayes' Rule

$$P(Y | X) = \frac{P(X | Y)P(Y)}{P(X)}$$

For two possible classes, ($Y=1$) and ($Y=0$):

$$P(Y = 1 | X) = \frac{P(X | Y = 1)P(Y = 1)}{P(X | Y = 1)P(Y = 1) + P(X | Y = 0)P(Y = 0)}.$$

- $Y=1$: you got flu
- $X=1$: you got coughed

Given:

$$P(Y = 1) = 0.05, \quad P(X = 1 | Y = 1) = 0.80, \quad P(X = 1 | Y = 0) = 0.20.$$

What is

$$P(\text{flu} | \text{cough}) = P(Y = 1 | X = 1) = ?$$



What does this has to do with function approximation?

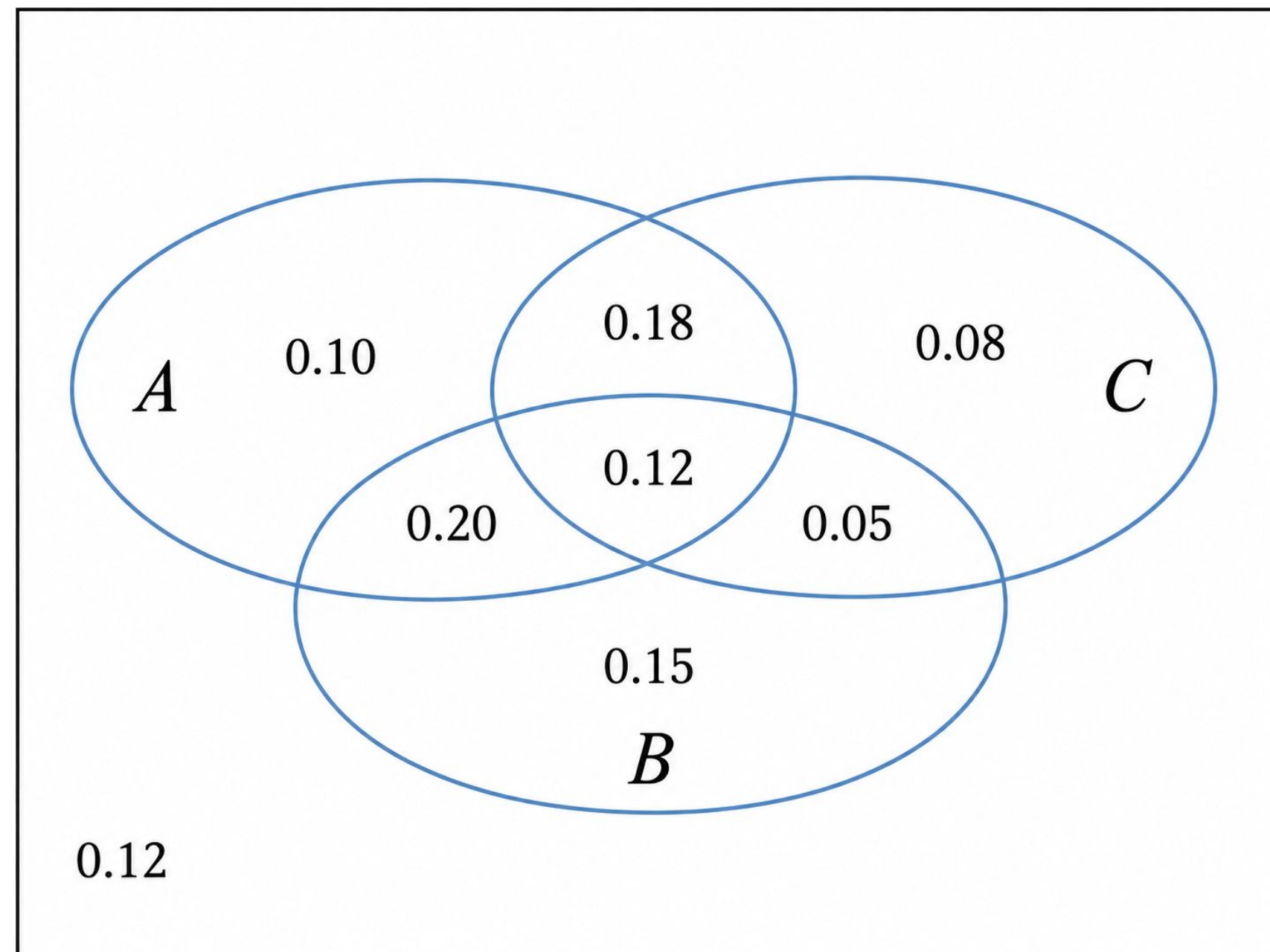
Instead of learning $F:X \rightarrow Y$

Learn $F(X) = \operatorname{argmax}_y P(Y=y \mid X)$

Joint Distribution

Recipe for making a joint distribution of M variables:

- Make a truth table listing all combinations of values (M Boolean variables $\rightarrow 2^M$ rows).
- For each combination of values, say how probable it is.
- By the axioms of probability, these probabilities must sum to 1.



A	B	C	Prob
0	0	0	0.12
0	0	1	0.08
0	1	0	0.15
0	1	1	0.05
1	0	0	0.10
1	0	1	0.18
1	1	0	0.20
1	1	1	0.12

Example: Boolean variables A,B,C

Using the Joint Distribution

Once we have the Joint Distribution, can ask for the probability of any logical expression involving these variables

$$P(E) = \sum_{\text{rows matching } E} P(\text{row})$$

$$P(C) = ?$$

$$P(C) = 0.08 + 0.05 + 0.18 + 0.12 = 0.43$$

A	B	C	Prob
0	0	0	0.12
0	0	1	0.08
0	1	0	0.15
0	1	1	0.05
1	0	0	0.10
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Example: Boolean variables A,B,C

Using the Joint Distribution

Once we have the Joint Distribution, can ask for the probability of any logical expression involving these variables

$$P(E) = \sum_{\text{rows matching } E} P(\text{row})$$

$$P(A \ \& \ C) = ?$$



$$P(A \ \& \ C) = 0.18 + 0.12 = 0.3$$

A	B	C	Prob
0	0	0	0.12
0	0	1	0.08
0	1	0	0.15
0	1	1	0.05
1	0	0	0.10
1	0	1	0.18
1	1	0	0.20
1	1	1	0.12

Example: Boolean variables A,B,C

Inference with the Joint Distribution

Once we have the Joint Distribution, can ask for the probability of any logical expression involving these variables

$$P(E_1 \mid E_2) = \frac{P(E_1 \wedge E_2)}{P(E_2)} = \frac{\sum_{\text{rows matching } E_1 \text{ and } E_2} P(\text{row})}{\sum_{\text{rows matching } E_2} P(\text{row})}$$

$$P(A \mid C) = ?$$



$$P(A \mid C) = 0.3 / 0.43 = 0.698$$

A	B	C	Prob
0	0	0	0.12
0	0	1	0.08
0	1	0	0.15
0	1	1	0.05
1	0	0	0.10
1	0	1	0.18
1	1	0	0.20
1	1	1	0.12

Example: Boolean variables A,B,C

Learning and the Joint Distribution

Suppose we want to learn the function

$$F:(A, B) \rightarrow C$$



$$F(A, B) = \operatorname{argmax}_c P(C=c \mid A, B)$$

One solution learn joint distribution from data, calculate $P(C \mid A, B)$

e.g., $P(C=1 \mid A=1, B=1) = ?$



A	B	C	Prob
0	0	0	0.12
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Example: Boolean variables A,B,C

Learning and the Joint Distribution

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One solution learn joint distribution from data, calculate $P(C \mid A, B)$

e.g., $P(C=1 \mid A=1, B=1) = ?$

$$\begin{aligned} P(C=1 \mid A=1, B=1) &= \frac{P(A=1, B=1, C=1)}{P(A=1, B=1)} \\ &= \frac{0.12}{0.20 + 0.12} \\ &= 0.375. \end{aligned}$$



A	B	C	Prob
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0	1	0	0.15
0	1	1	0.05
1	0	0	0.10
1	0	1	0.18
1	1	0	0.20
1	1	1	0.12

Example: Boolean variables A,B,C

Idea: learn classifiers by learning $P(Y|X)$

Consider:

- $Y=C$,
- $X = (A, B)$

A	B	$P(C=1 A,B)$	$P(C=0 A,B)$
0	0	0.4	0.6
0	1	0.25	0.75
1	0	0.643	0.357
1	1	0.375	0.625

A	B	C	Prob
0	0	0	0.12
0	0	1	0.08
0	1	0	0.15
0	1	1	0.05
1	0	0	0.10
1	0	1	0.18
1	1	0	0.20
1	1	1	0.12

Example: Boolean variables A,B,C

One approach: use this representation to learn $P(Y|X)$.

Are we done?!?



A Direct Conditional Table Does Not Scale

One approach: use this representation to learn $P(Y|X)$.

Main problem: learning $P(Y|X)$ might require more data than we have...

Number of features d	Possible feature vectors 2^d
2	4
10	1,024
20	1,048,576
100	$2^{100} \approx 1.27 \times 10^{30}$

Most feature combinations will have few or no training examples.
We need a more compact representation.

How Can We Make This Learnable?

1. Be smart about how to represent joint distributions

2. Be smart about how to estimate probabilities

Be smart about how to represent joint distributions

The Naïve Bayes Assumption

$$P(Y \mid X_1, \dots, X_d) \propto P(Y)P(X_1, \dots, X_d \mid Y).$$

Directly modeling $P(X_1, \dots, X_d \mid Y)$ requires probabilities for many feature combinations.

The Naïve Bayes algorithm assumes

$$P(X_1 = x_1, \dots, X_d = x_d \mid Y = y) = \prod_{j=1}^d P(X_j = x_j \mid Y = y)$$

Given the class Y , the features $X_1, \dots, X_d$ are assumed to be **mutually conditionally independent**.

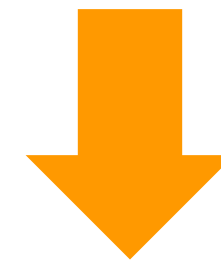
Conditional Independence: What Does It Mean?

$$X_1 \perp X_2 \mid Y$$

This means that after the class Y is known, observing X_2 provides no additional information about X_1 :

$$P(X_1 = x_1 \mid X_2 = x_2, Y = y) = P(X_1 = x_1 \mid Y = y).$$

$$P(X_1, X_2 \mid Y) = P(X_1 \mid Y)P(X_2 \mid Y).$$



If $X_1, \dots, X_n$ are mutually conditionally independent given Y , then

$$P(X_1, \dots, X_n \mid Y) = \prod_{i=1}^n P(X_i \mid Y)$$

The Naïve Bayes Assumption

If $X_1, \dots, X_n, Y$ are all Boolean, how many parameters do we need to describe $P(X_1, X_2, \dots, X_n|Y)$ and $P(Y)$?

- Without the conditional independence assumption: $2(2^n - 1) + 1$
- With conditional independence assumption: $2n + 1$

n	Without assumption	With assumption
10	2,047	21
100	$\approx 2.5 \times 10^{30}$	201

Naive Bayes on the ABC Example

Using Naïve Bayes, calculate $P(C \mid A, B)$

e.g., $P_{NB}(C=1 \mid A=1, B=1) = ?$



Bayes' Rule

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}$$

For two possible classes, ($Y=1$) and ($Y=0$):

$$P(Y = 1 \mid X) = \frac{P(X \mid Y = 1)P(Y = 1)}{P(X \mid Y = 1)P(Y = 1) + P(X \mid Y = 0)P(Y = 0)}.$$

$$P(X_1, X_2 \mid Y) = P(X_1 \mid Y)P(X_2 \mid Y).$$

A	B	C	Prob
0	0	0	0.12
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Example: Boolean variables A,B,C

Learning and the Joint Distribution

Suppose we want to learn the function $F(A, B) = \operatorname{argmax}_c P(C=c \mid A, B)$

Using Naïve Bayes, calculate $P(C \mid A, B)$

e.g., $P(C=1 \mid A=1, B=1) = ?$

$$P(C = 1) = 0.43, \quad P(C = 0) = 0.57.$$

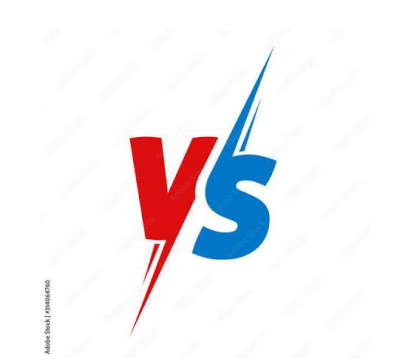
For $C = 1$:

$$P(A = 1 \mid C = 1) = \frac{0.18 + 0.12}{0.43} = \frac{0.30}{0.43},$$
$$P(B = 1 \mid C = 1) = \frac{0.05 + 0.12}{0.43} = \frac{0.17}{0.43}.$$

For $C = 0$:

$$P(A = 1 \mid C = 0) = \frac{0.30}{0.57},$$
$$P(B = 1 \mid C = 0) = \frac{0.35}{0.57}.$$

→ $P_{NB}(C = 1 \mid A = 1, B = 1) = \frac{0.1186}{0.1186 + 0.1842} \approx 0.392.$



$$P(C = 1 \mid A = 1, B = 1) = \frac{0.12}{0.12 + 0.20} = 0.375.$$



A	B	C	Prob
0	0	0	0.12
0	0	1	0.08
0	1	0	0.15
0	1	1	0.05
1	0	0	0.10
1	0	1	0.18
1	1	0	0.20
1	1	1	0.12

Example: Boolean variables A,B,C

Bayes Classification with Multiple Features

Let the observed feature vector be

$$\mathbf{x} = (x_1, x_2, \dots, x_d).$$

For each possible class (y_k), Bayes' rule gives

$$P(Y = y_k \mid \mathbf{X} = \mathbf{x}) = \frac{P(Y = y_k)P(\mathbf{X} = \mathbf{x} \mid Y = y_k)}{P(\mathbf{X} = \mathbf{x})}.$$

The numerator contains

$$\underbrace{P(Y = y_k)}_{\text{class prior}} \quad \underbrace{P(\mathbf{X} = \mathbf{x} \mid Y = y_k)}_{\text{joint feature likelihood}}.$$

Because $P(\mathbf{X} = \mathbf{x})$ is the same for every candidate class,

$$\hat{y} = \arg \max_{y_k} P(Y = y_k)P(\mathbf{X} = \mathbf{x} \mid Y = y_k)$$

Remaining problem:

How can we estimate the joint feature likelihood when $\mathbf{X}$ contains many features?

Naïve Bayes in a Nutshell

Naive Bayes assumes that the features are mutually conditionally independent given the class:

$$P(\mathbf{X} = \mathbf{x} \mid Y = y_k) = \prod_{i=1}^d P(X_i = x_i \mid Y = y_k).$$

For each class y_k , calculate the Naive Bayes score:

$$s_k(\mathbf{x}) = P(Y = y_k) \prod_{i=1}^d P(X_i = x_i \mid Y = y_k).$$

Predict the class with the largest score:

$$\hat{y} = \arg \max_{y_k} P(Y = y_k) \prod_{i=1}^d P(X_i = x_i \mid Y = y_k)$$

If posterior probabilities are needed, normalize the scores:

$$P(Y = y_k \mid \mathbf{X} = \mathbf{x}) = \frac{s_k(\mathbf{x})}{\sum_j s_j(\mathbf{x})}.$$

Bayes classifier + conditional independence = Naive Bayes.

Be smart about how to estimate probabilities

Naïve Bayes: Training and Prediction



How to estimate class priors and conditional feature probabilities?

Training Phase

Using only the training data, estimate the class priors:

$$\hat{\pi}_k = \hat{P}(Y = y_k).$$

Estimate the conditional feature probabilities:

$$\hat{\theta}_{i,v,k} = \hat{P}(X_i = v \mid Y = y_k).$$

Prediction Phase

For a new observation

$$\mathbf{x} = (x_1, x_2, \dots, x_d),$$

calculate a score for each class:

$$s_k(\mathbf{x}) = \hat{\pi}_k \prod_{i=1}^d \hat{P}(X_i = x_i \mid Y = y_k).$$

Predict the class with the largest score:

$$\hat{y} = \arg \max_{y_k} s_k(\mathbf{x})$$

The scores do not need to be normalized when only the predicted class is required.

Training data → estimated probabilities → class scores → prediction.

MLE for Discrete Naïve Bayes

Maximum likelihood estimation selects the parameters that maximize the likelihood of the training data:

$$(\hat{\pi}, \hat{\theta})_{\text{MLE}} = \arg \max_{\pi, \theta} P(D \mid \pi, \theta).$$

Let:

- N : total number of training examples
- N_k : number of examples with $Y = y_k$
- $N_{i,v,k}$: number of examples with $X_i = v$ and $Y = y_k$

Class-Prior Estimate

$$\hat{\pi}_k^{\text{MLE}} = \hat{P}(Y = y_k) = \frac{N_k}{N}$$

Conditional Feature Estimate

$$\hat{\theta}_{i,v,k}^{\text{MLE}} = \hat{P}(X_i = v \mid Y = y_k) = \frac{N_{i,v,k}}{N_k}$$

Limitation 1: Features May Be Conditionally Dependent

In real data, features may remain dependent even after conditioning on the class:

$$P(X_1, \dots, X_d \mid Y) \neq \prod_{i=1}^d P(X_i \mid Y).$$

The estimated posterior may differ from the true posterior:

$$P_{\text{NB}}(Y \mid X) \neq P(Y \mid X).$$

However, the predicted class can still be correct:

$$\arg \max_y P_{\text{NB}}(Y = y \mid X) = \arg \max_y P(Y = y \mid X).$$

- Posterior probabilities may be inaccurate or overconfident.
- The predicted class may still be correct.
- Good performance is not guaranteed and must be evaluated on held-out data.

Limitation 2: The Zero-Probability Problem

Suppose a feature value (v) is never observed within class (y_k):

$$N_{i,v,k} = 0.$$

The MLE estimate is therefore

$$\hat{P}_{\text{MLE}}(X_i = v \mid Y = y_k) = \frac{N_{i,v,k}}{N_k} = 0.$$

The Naive Bayes score for class (y_k) is

$$s_k(\mathbf{x}) = \hat{P}(Y = y_k) \prod_{i=1}^d \hat{P}(X_i = x_i \mid Y = y_k).$$

If any one of these probabilities is zero, then

$$s_k(\mathbf{x}) = 0$$

regardless of the values of the other features.

Not observed in the training data does not necessarily mean impossible.

Additive Smoothing

For feature X_i with J_i possible values, additive smoothing estimates

$$\hat{P}_\lambda(X_i = v \mid Y = y_k) = \frac{N_{i,v,k} + \lambda}{N_k + \lambda J_i}$$

where $\lambda > 0$ is the smoothing parameter.

- Each possible value receives a pseudo-count of λ .
- Unseen values receive a small nonzero probability.
- Values with large observed counts change only slightly.

When $\lambda = 1$, we obtain **Laplace smoothing**, also called **add-one smoothing**:

$$\hat{P}_{\text{Laplace}}(X_i = v \mid Y = y_k) = \frac{N_{i,v,k} + 1}{N_k + J_i}.$$



This formula looks familiar

Add-One Smoothing: Example

Suppose that among 1,000 legitimate emails, the feature topic has three possible values:

Topic	Count
Promotion	0
Work	990
Account	10

Using MLE:

$$P_{\text{MLE}}(\text{promotion} \mid \text{legitimate}) = \frac{0}{1000} = 0.$$

Using add-one smoothing:

$$P(\text{promotion} \mid \text{legitimate}) = \frac{0 + 1}{1000 + 3} = \frac{1}{1003},$$

$$P(\text{work} \mid \text{legitimate}) = \frac{990 + 1}{1000 + 3} = \frac{991}{1003},$$

$$P(\text{account} \mid \text{legitimate}) = \frac{10 + 1}{1000 + 3} = \frac{11}{1003}.$$

The smoothed probabilities still sum to one:

$$\frac{1 + 991 + 11}{1003} = 1.$$

What is this?



MAP Interpretation of Additive Smoothing

For a categorical variable with J possible values, place a symmetric Dirichlet prior on its probability parameters:

$$\boldsymbol{\theta} \sim \text{Dirichlet}(\alpha, \dots, \alpha).$$

The MAP estimate is

$$\hat{\theta}_{j,\text{MAP}} = \frac{N_j + \alpha - 1}{N + J(\alpha - 1)}$$

Set $\alpha = 2$:

$$\begin{aligned}\hat{\theta}_{j,\text{MAP}} &= \frac{N_j + 2 - 1}{N + J(2 - 1)} \\ &= \frac{N_j + 1}{N + J}.\end{aligned}$$

Therefore,

Add-one smoothing = MAP under a symmetric $\text{Dirichlet}(2, \dots, 2)$ prior.

For a binary feature, the corresponding prior is

$$\theta \sim \text{Beta}(2, 2),$$

and

$$\hat{\theta}_{\text{MAP}} = \frac{H + 1}{H + T + 2}.$$



Practice: Naïve Bayes Email Classification

Consider the following new email:

$\mathbf{x} = (\text{sender known=no, topic=work, contains link=no, urgent words=no})$

1. Predict its class using MLE estimates.
2. Identify any zero-probability problem.
3. Repeat the prediction using add-one smoothing.



ID	Sender known	Topic	Contains link	Urgent words	Class
1	no	promotion	yes	yes	spam
2	no	promotion	yes	no	spam
3	no	account	yes	yes	spam
4	no	work	yes	yes	spam
5	yes	promotion	yes	no	spam
6	no	account	no	yes	spam
7	yes	account	yes	yes	spam
8	yes	work	no	no	legitimate
9	yes	account	no	no	legitimate
10	yes	account	yes	no	legitimate
11	yes	work	yes	no	legitimate
12	yes	account	no	yes	legitimate
13	yes	work	no	yes	legitimate
14	yes	account	no	no	legitimate

Naïve Bayes Classifier: Strengths and Limitations

Strengths	Limitations
Fast to train and predict	Independence assumption may be unrealistic
Works well with sparse, high-dimensional data	Zero counts require smoothing
Requires relatively little training data	Estimated probabilities may be poorly calibrated

What you should know...

- What's the The Naïve Bayes Assumption
- Why we use it
- How do we learn it
- Why is Bayesian estimation important